

VENKATESH PATSAMATLA

Senior Cloud Data Engineer

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PROFILE SUMMARY

Strategic and results-driven **Data Engineer** with **7+ years of experience** architecting **scalable data platforms**, **real-time streaming pipelines**, and **cloud-native analytics** solutions across **Healthcare, Finance, and Big Tech** environments. Expert in building **ETL/ELT** workflows using **Python, SQL, Apache Spark, Databricks**, and **Kafka**, leveraging modern cloud ecosystems including **Azure (ADF, ADLS, Synapse, AKS)**, **AWS (S3, Glue, Lambda, Redshift)**, and **Snowflake**. Strong background in **data modeling**, **Delta Lake**, **Data Lakehouse**, **distributed computing**, and **CI/CD automation** with **Terraform, GitHub Actions**, and **Azure DevOps**. Skilled in partnering with **Data Science** and **Product** teams to drive **data governance, security**, and **performance optimization** that reduce compute costs and accelerate **AI/ML readiness**. Proven success enabling **advanced analytics**, improving **data quality & reliability**, and delivering **business-impacting insights** at enterprise scale.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, Scala, Java, Bash

Big Data & Distributed Processing: Apache Spark, Hadoop, Hive, HDFS, Sqoop, MapReduce, Kafka

Cloud Platforms: AWS (S3, Glue, Lambda, Redshift, EMR, EC2), Azure (Data Factory, Synapse, Databricks), GCP (BigQuery, Cloud Storage)

Data Engineering, ETL & Automation: Databricks (PySpark), Apache Kafka, Azure Data Factory, AWS Glue, dbt, Alteryx, Apache NiFi (optional addition if you touched it — if not, I'll remove), Unix/Linux Shell Scripting

Data Warehousing & Databases: Snowflake, Amazon Redshift, Google BigQuery, Azure Synapse, PostgreSQL, MySQL, MongoDB, Cassandra

Orchestration & CI/CD: Apache Airflow, Git, GitHub Actions, Jenkins (optional), Azure DevOps, Maven

Infrastructure & DevOps: Docker, Kubernetes (AKS), Terraform, CI/CD Pipelines

Monitoring & Logging: Azure Log Analytics, CloudWatch

Data Modeling & Governance: Star & Snowflake Schema, Dimensional Modeling (Kimball), Data Quality, Data Lineage, Role-Based Access Control (RBAC), Metadata Management

BI & Reporting: Power BI, Amazon QuickSight

ORGANISATIONAL EXPERIENCE

Senior Data Engineer | Charles Schwab | June 2024 – April 2025 | USA

Tech Stack: Azure (ADF, ADLS, Databricks, AKS, Key Vault), Kafka, Power BI, dbt, GitHub Actions, Terraform

- Architecting enterprise-scale **data modernization initiative**, replacing on-prem data marts with **Azure Databricks + Delta Lakehouse**, improving pipeline performance by **45%** and lowering compute cost by **25% YoY**
- Designed a **real-time risk analytics pipeline** using **Kafka → Event Hub → ADLS → Databricks**, delivering market data feeds with **sub-5s latency**, enabling faster fraud detection and portfolio risk adjustments
- Established **governance-first data ingestion framework** with **RBAC, Key Vault encryption, PII masking**, ensuring **SOX & FINRA compliance** across ~50 critical datasets
- Led development of **dbt-enabled transformation layer**, reducing manual SQL work by **60%** and driving higher data trust (95%+ DQ success rate)
- Built **CI/CD workflows** for Databricks and ADF using GitHub Actions and Terraform, reducing deployment effort from **4 hours → 20 minutes**
- Actively mentor 3 engineers & enforce **best practices in performance optimization, error recovery architecture, and automated lineage tracking**

Data Engineer | Meta | Jan 2023 – May 2024 | USA

Tech Stack: Databricks, Spark, Snowflake, Kafka, Airflow, Python, AWS S3, dbt

- Designed **petabyte-scale** ETL pipelines supporting Meta's product analytics teams; reduced data refresh time by **35%** using Spark partitioning strategies & cost-aware cluster tuning
- Built **high-frequency Kafka streaming** framework capturing >2B daily user events with **>99.98% uptime**, fueling real-time recommendations across apps
- Integrated **dbt transformation layer** into Snowflake, reducing upstream data issues by **60%** and improving BI SLA from **88% → 99.7%**
- Partnered with ML engineers to deliver **feature-store ready datasets** that accelerated experimentation cycle by **20%** per model release

Data Engineer | Accenture | Sept 2018 – Nov 2021 | India

Tech Stack: Azure ADF, ADLS, Azure Synapse, Databricks, Power BI, Python, SQL

- Migrated multiple enterprise data applications to **Azure Synapse + ADLS**, providing **3–5x faster** query performance and **30% reduced storage cost**

- Engineered **CDC-driven ETL** using ADF + Functions, reducing pipeline durations from **6 hours to 1.5 hours**
- Automated business rules validation using **PySpark and SQL**, reducing QA effort by **70%** and boosting data reliability
- Developed **role-based access** and **data retention** policies to ensure **GDPR & HIPAA compliance** for healthcare and retail clients
- Delivered curated data models for **Power BI** reporting covering finance, product, and operations KPIs, improving adoption by **40%**

Data Engineer | Legato Health | May 2017 – Aug 2018 | India

Tech Stack: Hadoop, Hive, HDFS, Sqoop, Shell, Oracle 11g

- Designed ETL frameworks to process healthcare **claims & EHR data** with **99.9% accuracy**, ensuring HIPAA compliance for protected health data
- Automated batch workflows using **Shell scripting**, reducing execution effort by **40%** and failures by **30%**
- Developed care-management reporting models that reduced care-gap detection time by **35%**
- Improved ingestion throughput by tuning Hive queries and storage formats, reducing memory consumption by **~20%**

EDUCATION

Master of Science – Big Data Analytics & Information Technology

University of Central Missouri | USA | Jan 2022 – May 2023

Relevant Coursework: Python, SQL, PL/SQL, Java, AWS, Spark, Data Management

Post-Graduate Diploma – Data Science

Manipal Academy of Higher Education | India | Jul 2019 – Sep 2020

Coursework: Big Data, Machine Learning, Computer Vision, NLP, Predictive Models

Bachelor of Engineering – Mechanical Engineering

Anna University | India | Jun 2013 – Apr 2017

CERTIFICATIONS

AWS Certified Solutions Architect – Associate | Credential ID: QGDSJ6H2Z14E1S5M | Nov 2022

Tableau Data Scientist Certification | May 2020

Tableau Analyst Certification | May 2020